

[EECS 304] Control I

Sheet (3) Steady State Analysis

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Outline

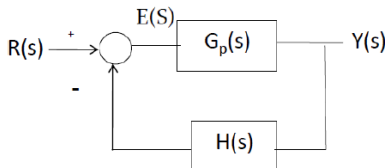
1 Overview

2 Problems

- Problem 1
- Problem 2
- Problem 3
- Problem 4
- Problem 5

Overview

- $Y(s) = R(s) \times T.F.$
- $E(s) = \frac{R(s)}{1 + G_p(s)H(s)}$



$$\therefore E(s) = R(s) - H(s)Y(s)$$

$$\therefore E(s) = R(s) - H(s)[R(s) * T.F.]$$

$$\therefore E(s) = R(s) - H(s)\left[R(s) * \frac{G_p(s)}{1 + G_p(s)H(s)}\right]$$

$$\therefore E(s) = R(s)\left[1 - \frac{G_p(s)H(s)}{1 + G_p(s)H(s)}\right]$$

$$\therefore E(s) = R(s)\left[\frac{1}{1 + G_p(s)H(s)}\right]$$

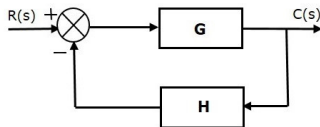
$$\therefore E(s) = \frac{R(s)}{1 + G_p(s)H(s)}$$

How to get e_{ss} ?

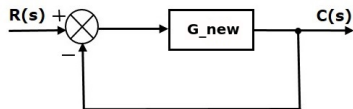
- If $E(s)$ is given/got $\rightarrow e_{ss} = \lim_{s \rightarrow 0} sE(s)$
- If the **actuating error** is required $\rightarrow$ Use the table below. (Regardless of $H(s)$)
- If the **absolute error** is required $\rightarrow$ The system must be of unity feedback first, then use the table below.

	$R(s)$	e_{ss}	Error Constant Value
e_{ss} due to step input	$\frac{M}{s}$	$\frac{M}{1 + K_p}$	$K_p = \lim_{s \rightarrow 0} G_p(s)H(s)$ position error constant
e_{ss} due to ramp input	$\frac{M}{s^2}$	$\frac{M}{K_v}$	$K_v = \lim_{s \rightarrow 0} sG_p(s)H(s)$ velocity error constant
e_{ss} due to parabolic input	$\frac{M}{s^3}$	$\frac{M}{K_a}$	$K_a = \lim_{s \rightarrow 0} s^2 G_p(s)H(s)$ acceleration error constant

Unity Feedback



$\rightarrow$



$$G_{\text{new}}(s) = \frac{G(s)}{1 + G(s)H(s) - G(s)}$$

System Type

- $G_p(s)H(s) = \frac{K(1 + \tau_1 s)(1 + \tau_2 s)\dots}{s^N(1 + \tau_a s)(1 + \tau_b s)\dots}$, where N is the type of the system

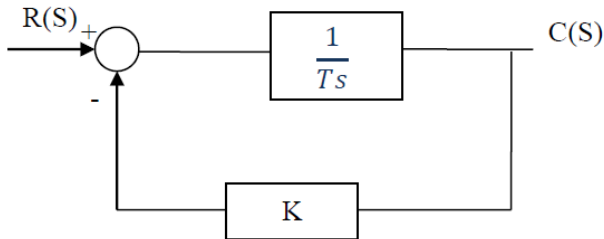
	Steady State Error		
	Step Input	Ramp Input	Parabolic Input
Type 0 system	$\frac{M}{1 + K}$	∞	∞
Type 1 system	0	$\frac{M}{K}$	∞
Type 2 system	0	0	$\frac{M}{K}$

Problem 1

Problem 1

For the following system find:

- ① The transfer function $C(s)/R(s)$.
- ② The output $c(t)$ if
 - a $R(s)$ is step input.
 - b $R(s)$ is ramp input.
 - c $R(s)$ is parabolic input.
- ③ The steady state output c_{ss} and the steady state error e_{ss} if possible.



Problem 1

$$\text{a) } TF = \frac{C(s)}{R(s)} = \frac{\frac{1}{Ts}}{1 + \frac{K}{Ts}} = \frac{1}{Ts + K}$$

$$\text{b) i) at } r(t) = u(t) \rightarrow R(s) = \frac{1}{s}$$

$$\therefore C(s) = \frac{1}{s(Ts + K)} = \frac{A}{s} + \frac{B}{Ts + K}$$

$$A = \frac{1}{K}, \quad B = \frac{-T}{K}$$

$$\therefore C(s) = \frac{1/K}{s} - \frac{T/K}{Ts + K} = \frac{1/K}{s} - \frac{1/K}{s + \frac{K}{T}}$$

$$\therefore c(t) = \left(\frac{1}{K} - \frac{1}{K} e^{-\frac{K}{T}t} \right) u(t)$$

Problem 1

b) ii) at $r(t) = t \rightarrow R(s) = \frac{1}{s^2}$

$$\therefore C(s) = \frac{1}{s^2(Ts + K)} = \frac{A}{s^2} + \frac{B}{s} + \frac{C}{Ts + K}$$

$$A = \frac{1}{K}, \quad C = \frac{T^2}{K^2}$$

For B:

$$1 = A(Ts + K) + Bs(Ts + K) + Cs^2$$

Compare coefficients:

$$0_{of\ s^2} = BT + C$$

$$\therefore B = \frac{-C}{T} = \frac{-T}{K^2}$$

$$\therefore C(s) = \frac{1/K}{s^2} - \frac{T/K^2}{s} + \frac{T^2/K^2}{Ts + K}$$

$$\therefore c(t) = \left(\frac{1}{K}t - \frac{T}{K^2} + \frac{T}{K^2}e^{\frac{-K}{T}t} \right) u(t)$$

Problem 1

b) iii) at $r(t) = \frac{t^2}{2} \rightarrow R(s) = \frac{1}{s^3}$

$$\therefore C(s) = \frac{1}{s^3(Ts + K)} = \frac{A}{s^3} + \frac{B}{s^2} + \frac{C}{s} + \frac{D}{Ts + K}$$

$$A = \frac{1}{K}, \quad D = \frac{-T^3}{K^3}$$

For B, C:

$$1 = A(Ts + K) + Bs(Ts + K) + Cs^2(Ts + K) + Ds^3$$

Compare coefficients:

$$0_{\text{of } s^3} = CT + D$$

$$\therefore C = \frac{-D}{T} = \frac{T^2}{K^3}$$

$$0_{\text{of } s^2} = BT + CK$$

$$\therefore B = \frac{-CK}{T} = \frac{-T^2}{K^3} K \frac{1}{T} = \frac{-T}{K^2}$$

Problem 1

$$\therefore C(s) = \frac{1/K}{s^3} - \frac{T/K^2}{s^2} + \frac{T^2/K^3}{s} - \frac{T^3/K^3}{Ts + K}$$

$$\therefore c(t) = \left(\frac{1}{K} \frac{t^2}{2} - \frac{T}{K^2} t + \frac{T^2}{K^3} - \frac{T^2}{K^3} e^{\frac{-K}{T} t} \right) u(t)$$

c) **For absolute error:**

$$\therefore H(s) \neq 1 \rightarrow G'(s) = \frac{G(s)}{1 + G(s)H(s) - G(s)}$$

$$\therefore G'(s) = \frac{\frac{1}{Ts}}{1 + \frac{K}{Ts} - \frac{1}{Ts}}$$

$$\therefore G'(s) = \frac{1}{Ts + (K - 1)}, K > 1$$

Problem 1

i) For $r(t) = u(t)$

$$c_{ss} = \lim_{s \rightarrow 0} sC(s) = \lim_{s \rightarrow 0} \frac{1}{Ts + K} = \frac{1}{K}$$

$$\therefore e_{ss}|_{unit\ step} = \frac{1}{1 + K_p}$$

$$K_p = \lim_{s \rightarrow 0} G'(s) = \lim_{s \rightarrow 0} \frac{1}{Ts + (K - 1)} = \frac{1}{K - 1}$$

$$\therefore e_{ss}|_{unit\ step} = \frac{1}{1 + K_p} = \frac{K - 1}{K}$$

ii) For $r(t) = t$

$$c_{ss} = \lim_{s \rightarrow 0} sC(s) = \lim_{s \rightarrow 0} \frac{1}{s(Ts + K)} = \infty$$

$$\therefore e_{ss}|_{unit\ ramp} = \frac{1}{K_v}$$

Problem 1

$$K_v = \lim_{s \rightarrow 0} sG'(s) = \lim_{s \rightarrow 0} \frac{s}{Ts + (K - 1)} = 0$$

$$\therefore e_{ss}|_{unit\ ramp} = \infty$$

iii) For $r(t) = \frac{t^2}{2}$

$$c_{ss} = \lim_{s \rightarrow 0} sC(s) = \lim_{s \rightarrow 0} \frac{1}{s^2(Ts + K)} = \infty$$

$$\therefore e_{ss}|_{parabolic} = \frac{1}{K_a}$$

$$K_a = \lim_{s \rightarrow 0} s^2 G'(s) = \lim_{s \rightarrow 0} \frac{s^2}{Ts + (K - 1)} = 0$$

$$\therefore e_{ss}|_{parabolic} = \infty$$

Problem 1

For actuating error:

i) For $r(t) = u(t)$

$$\therefore e_{ss}|_{unit\ step} = \frac{1}{1 + K_p}$$

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{K}{Ts} = \infty$$

$$\therefore e_{ss}|_{unit\ step} = \frac{1}{1 + K_p} = 0$$

ii) For $r(t) = t$

$$\therefore e_{ss}|_{unit\ ramp} = \frac{1}{K_v}$$

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s) = \lim_{s \rightarrow 0} s \frac{K}{Ts} = \frac{K}{T}$$

$$\therefore e_{ss}|_{unit\ ramp} = \frac{T}{K}$$

Problem 1

iii) For $r(t) = \frac{t^2}{2}$

$$\therefore e_{ss}|_{parabolic} = \frac{1}{K_a}$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) = \lim_{s \rightarrow 0} s^2 \frac{K}{Ts} = 0$$

$$\therefore e_{ss}|_{parabolic} = \infty$$

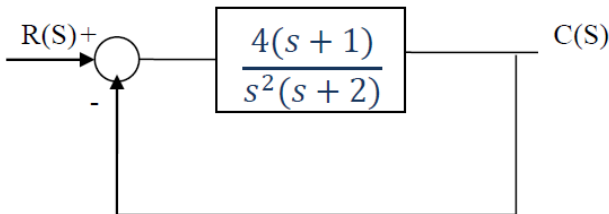
Problem 2

Problem 2

For the following system find:

- 1 The position, velocity and acceleration error constants.
- 2 The steady state error when the input is

$$R(s) = \frac{3}{s} - \frac{1}{s^2} + \frac{1}{2s^3}$$



Problem 2

a)

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \lim_{s \rightarrow 0} \frac{4(s+1)}{s^2(s+2)} = \infty$$

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s) = \lim_{s \rightarrow 0} \frac{4(s+1)}{s(s+2)} = \infty$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) = \lim_{s \rightarrow 0} \frac{4(s+1)}{(s+2)} = 2$$

b) For $R(s) = \frac{3}{s} - \frac{1}{s^2} + \frac{1}{2s^3}$

$$\therefore e_{ss}|_{total} = \frac{M_1}{1 + K_p} + \frac{M_2}{K_v} + \frac{M_3}{K_a}$$

$$M_1 = 3, \quad M_2 = -1, \quad M_3 = \frac{1}{2}$$

$$\therefore e_{ss}|_{total} = 0 + 0 + \frac{1}{2} * \frac{1}{2} = 0.25$$

Problem 3

Problem 3

Find the transfer function for the following systems, then get the steady state value of the output for input $r(t) = 1 + 2t$:

① $\ddot{y} + 3\ddot{y} + \dot{y} - 2y = r(t)$

② $y^{(4)} + 2\ddot{y} + 5\ddot{y} + \dot{y} + 2y = r(t)$

① $\ddot{y} + 3\ddot{y} + \dot{y} - 2y = r(t)$

Take Laplace for the two sides:

$$s^3 Y(s) + 3s^2 Y(s) + sY(s) - 2Y(s) = R(s)$$

$$\therefore TF = \frac{Y(s)}{R(s)} = \frac{1}{s^3 + 3s^2 + s - 2}$$

This system isn't stable as the coefficient of s^0 in the characteristic equation is negative $\rightarrow$ steady state value doesn't exist.

Problem 3

$$② \quad y^{(4)} + 2\ddot{y} + 5\ddot{y} + \dot{y} + 2y = r(t)$$

Take Laplace for the two sides:

$$s^4 Y(s) + 2s^3 Y(s) + 5s^2 Y(s) + sY(s) + 2Y(s) = R(s)$$

$$\therefore TF = \frac{Y(s)}{R(s)} = \frac{1}{s^4 + 2s^3 + 5s^2 + s + 2}$$

$$\therefore Y(s) = \frac{R(s)}{s^4 + 2s^3 + 5s^2 + s + 2} = \frac{1/s + 2/s^2}{s^4 + 2s^3 + 5s^2 + s + 2}$$

$$\therefore y_{ss} = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} \frac{1/s + 2/s^2}{s^4 + 2s^3 + 5s^2 + s + 2} = \infty$$

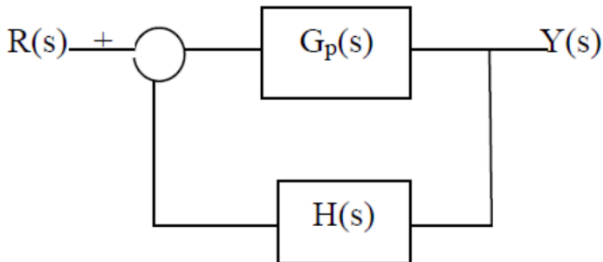
Problem 4

Problem 4

A feedback system has the following transfer functions:

$$G_p(s) = \frac{K}{(Ts + 1)}, \quad H(s) = \frac{1}{s}$$

Determine the static error coefficients for this system.



Problem 4

$$K_p = \lim_{s \rightarrow 0} G_p(s)H(s) = \lim_{s \rightarrow 0} \frac{K}{s(Ts + 1)} = \infty$$

$$K_v = \lim_{s \rightarrow 0} sG_p(s)H(s) = \lim_{s \rightarrow 0} \frac{K}{Ts + 1} = K$$

$$K_a = \lim_{s \rightarrow 0} s^2 G_p(s)H(s) = \lim_{s \rightarrow 0} \frac{Ks}{Ts + 1} = 0$$

Problem 5

Problem 5

A unity feedback system has a forward transfer function of:

$$G_p(s) = \frac{12(s + 4)}{s(s + 1)(s + 3)(s^2 + 2s + 10)}$$

- 1 Determine the static error coefficients for this system.
- 2 Determine the steady state error and the steady state output for a reference input $r(t) = 16 + 2t$ and for $r(t) = 5t^2$.

Problem 5

$\therefore$ Unity Feedback $\rightarrow H(s) = 1$

1

$$K_p = \lim_{s \rightarrow 0} G_p(s) = \lim_{s \rightarrow 0} \frac{12(s+4)}{s(s+1)(s+3)(s^2+2s+10)} = \infty$$

$$K_v = \lim_{s \rightarrow 0} sG_p(s) = \lim_{s \rightarrow 0} \frac{12(s+4)}{(s+1)(s+3)(s^2+2s+10)} = 1.6$$

$$K_a = \lim_{s \rightarrow 0} s^2 G_p(s) = \lim_{s \rightarrow 0} s \frac{12(s+4)}{(s+1)(s+3)(s^2+2s+10)} = 0$$

Problem 5

② For $r(t) = 16 + 2t \rightarrow R(s) = \frac{16}{s} + \frac{2}{s^2}$

$$\therefore e_{ss}|_{total} = \frac{16}{1 + K_p} + \frac{2}{K_v}$$

$$\therefore e_{ss}|_{total} = 0 + \frac{2}{1.6} = 1.25$$

$$\therefore y_{ss} = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s(R(s) * T.F.)$$

$$\therefore T.F. = \frac{12(s + 4)}{s(s + 1)(s + 3)(s^2 + 2s + 10) + 12(s + 4)}$$

$$\therefore y_{ss} = \lim_{s \rightarrow 0} s \left(\frac{16}{s} + \frac{2}{s^2} \right) \left(\frac{12(s + 4)}{s(s + 1)(s + 3)(s^2 + 2s + 10) + 12(s + 4)} \right)$$

$$\therefore y_{ss} = \infty$$

Problem 5

2 For $r(t) = 5t^2 \rightarrow R(s) = \frac{10}{s^3}$

$$\therefore e_{ss}|_{total} = \frac{10}{K_a}$$

$$\therefore e_{ss}|_{total} = \frac{10}{0} = \infty$$

$$\therefore y_{ss} = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} s(R(s) * T.F.)$$

$$\therefore T.F. = \frac{12(s+4)}{s(s+1)(s+3)(s^2+2s+10)+12(s+4)}$$

$$\therefore y_{ss} = \lim_{s \rightarrow 0} s \left(\frac{10}{s^3} \right) \left(\frac{12(s+4)}{s(s+1)(s+3)(s^2+2s+10)+12(s+4)} \right)$$

$$\therefore y_{ss} = \infty$$

Thank you!

Questions?

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